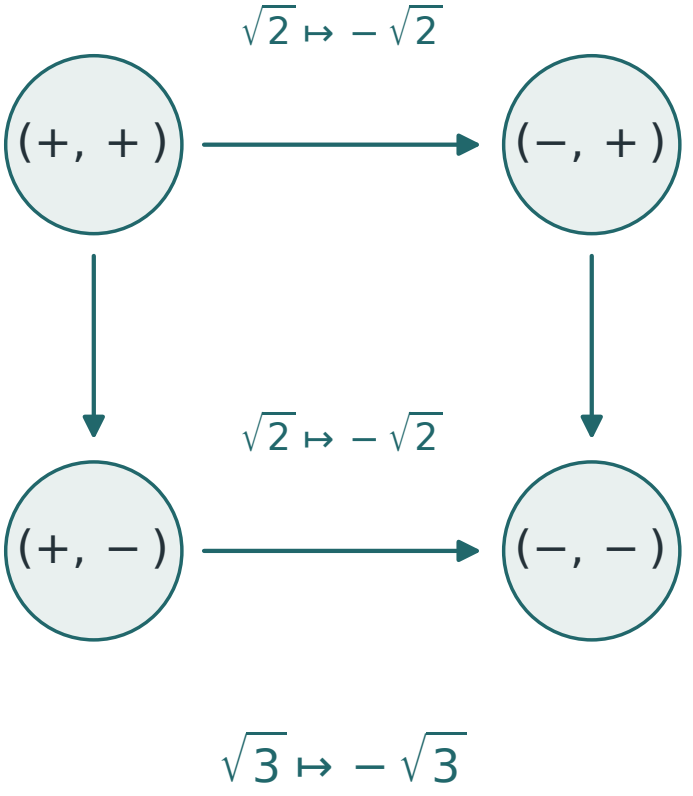
Two independent sign changes

WRITTEN PROOF / the biquadratic field



$$L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$$

$$\mathrm{Gal}(L/\mathbb{Q}) \cong C_2 \times C_2$$

- $\mathbb{Q}(\sqrt{2})$
- $\mathbb{Q}(\sqrt{3})$
- $\mathbb{Q}(\sqrt{6})$

The three order-two subgroups give exactly three quadratic intermediate fields.